

**Phase 1 — Client Proposal**  
IAT461 - Summer 2026 - Stin (S.D)

**Entity:** Job Accommodation Network (JAN)

(Dataset is fictional, company is real but problem definition is made up)

The leading national service providing free, expert guidance on workplace accommodations, and disability employment legislation. JAN assists thousands of employers and HR specialists each year in evaluating accommodation inquiries across health conditions.

**Business Context & Strategic Options:** Workplace accommodation inquiries have surged, and internal consultation logs show significant variance in approval rates and fulfillment delays across different health conditions. Management is debating three options for targeting their technical assistance and advocacy resources:

- **Option A:** Uniformly expand HR training programs across all employer departments and disability categories to increase overall accommodation awareness.
- **Option B:** Restrict technical assistance and legal support exclusively to high-cost physical workplace modifications (e.g., ramps and architectural changes), assuming financial cost is the primary barrier to accommodation approval.
- **Option C:** Audit accommodation decisions to identify specific, less visible disability categories and HR reviewers experiencing disproportionate denial rates or implementation friction.

**The Task:**

- **Phase 1 (EDA & Strategy Validation):** Using the provided dataset, conduct an exploratory analysis of accommodation costs, approval rates, and processing delays to evaluate Options A, B, and C. Identify which option the data justifies pursuing by evaluating whether non-cost factors (like disability type or HR reviewer) drive denials.
- **Phase 2 (Data Science Implementation):** Assuming management proceeds with Option C, build machine learning models to predict request approval status (`approval_status`) and fulfillment delays (`days_to_implement`). Use these predictions to pinpoint which health categories, severe request types, or specific HR reviewers exhibit systemic bias, and evaluate how fulfillment friction impacts long-term worker retention (`retained_6mo`).

**Dataset Provided:** `workplace_accommodation_synthetic.csv`

- <https://github.com/cda79/IAT461-Final/tree/main/proposal> (See README.md)
- 260 structured request records including disability types, accommodation categories, estimated costs, functional severity levels, HR reviewer IDs, approval outcomes, implementation wait times, and 6-month retention flags

| Column Name             | Data Type | Description & Purpose       |
|-------------------------|-----------|-----------------------------|
| <code>request_id</code> | String    | Identifier for each request |

|                        |                                                         |                                                                                             |
|------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------|
| disability_category    | Categorical                                             | Health classification                                                                       |
| accommodation_category | Categorical                                             | Type of adjustment requested                                                                |
| employee_tenure_years  | Float                                                   | Length of employee service in years.                                                        |
| hourly_wage_usd        | Float                                                   | Hourly wage of the requesting employee.                                                     |
| has_college_degree     | Binary (0/1)<br>1 = Degree held<br>0 = No degree        | Highest education indicator                                                                 |
| is_cost_incurred       | Binary (0/1)<br>1 = Direct cost involved<br>0 = No cost | Financial expense indicator                                                                 |
| accommodation_cost_usd | Float                                                   | Direct financial cost in USD                                                                |
| request_severity_level | Ordinal                                                 | Self-reported functional necessity                                                          |
| remote_work_status     | Categorical                                             | Working arrangement                                                                         |
| department             | Categorical                                             | Employer business unit                                                                      |
| hr_reviewer_id         | Categorical                                             | ID of the evaluating HR representative                                                      |
| approval_status        | Categorical                                             | Primary decision outcome                                                                    |
| denial_reason          | Categorical                                             | Stated rationale if denied                                                                  |
| days_to_implement      | Numeric                                                 | Calendar days from request submission to fulfillment.<br>Missing (NaN) for denied requests. |
| retained_6mo           | Binary (0/1)<br>1 = Retained<br>0 = Resigned/Terminated | Whether the employee remained 6 months later                                                |

**Success Criteria:** An exploratory data analysis demonstrating why uniform training or cost-only focus (Options A and B) are unfeasible, followed by a predictive model that reliably pinpoints which health conditions, severe request types, or specific HR reviewers experience disproportionate denial rates and implementation delays to guide JAN's advocacy budget and training modules.